

☆ Fwd: Seminar on Enabling Resource-Efficient AI via SW-HW Co-design

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Please see the talk announcement below:

----- Forwarded message -----

From: **Amit Trivedi** <amitr@uic.edu>

Date: Tue, Aug 12, 2025 at 11:56 AM

Subject: Seminar on Enabling Resource-Efficient AI via SW-HW Co-design

To: <COE-FACULTY@listserv.uic.edu>

Dear Colleague,

Professor Jong Hwan Ko from Sungkyunkwan University (SKKU) and his student cohort will be visiting us this Thursday and giving a seminar on Resource-Efficient AI via SW-HW Co-design. The seminar addresses the challenge of deploying increasingly large and complex AI models on resource-constrained systems through principled software-hardware co-design. Their group's work has received spotlight recognition in top AI conferences such as CVPR. Please see the talk details below. I would really appreciate it if you could share this with your students. If you are able to attend, I believe it will be a very interesting and insightful talk. Thanks again, and hope to see you there.

Date and Time: Aug 14, 2025, 4 PM – 6 PM**Meeting Room: ERF 1047****Zoom Link:** <https://uic.zoom.us/j/amitr>

The rapid growth of AI model size and complexity demands efficient deployment on resource-constrained systems. This challenge calls for a principled co-design of SW and HW, as conventional approaches often fail to exploit the joint optimization opportunities. In this seminar, we explore a range of SW-HW co-design techniques geared towards resource-efficient AI across a wide range of real-world applications. First, we present a series of model compression techniques, with a focus on mixed-precision and multi-precision quantization to accelerate both inference and training. Next, we explore how compression techniques can be co-optimized with next-generation computing paradigms (e.g., in-memory computing) to enhance computational efficiency. Through this seminar, we aim to offer practical pathways for efficient AI deployment under resource constraints.

Speaker: Jong Hwan Ko, Kang Eun Jeon, Johnny Rhe, Sangheum Yeon**Jong Hwan Ko****Kang Eun Jeon****Johnny Rhe****Sangheum Yeon**

Jong Hwan Ko received the dual B.S. degrees in computer science and engineering and mechanical and aerospace engineering and the M.S. degree in electrical engineering and computer science from Seoul National University, and the Ph.D. degree from the School of Electrical and Computer Engineering, Georgia Tech, in 2018. During his seven years research experience at the Agency for Defense Development (ADD) in South Korea, he conducted advanced research on the design and performance analysis of military wireless sensor networks. He is currently an Associate Professor at Sungkyunkwan University (SKKU), South Korea, where his research focuses on deep learning model compression and accelerator design for on-device AI, and low-power image sensor systems and multi-modal AI processing. He has received the Best Paper Award from the International Symposium on Low Power Electronics and Design (ISLPED), in 2016.

Kang Eun Jeon is a postdoctoral researcher at SKKU, advised by Prof Jong Hwan Ko. He earned his PhD from HKUST under the supervision of Prof James She and Prof Jun Zhang, where he developed luXbeacon, an open-sourced energy harvesting IoT device. luXbeacon was deployed by over 10 industry and academic partners, including Hong Kong government agencies such as EMSD and OGCI, for smart indoor navigation and COVID alert applications. His recent research focuses on efficient AI systems for power- and compute-constrained edge devices.

Johnny Rhe received the B.S. Degree in Electrical Engineering from Sungkyunkwan University, South Korea, in 2019, where he is currently pursuing the Ph.D. degree in Electrical and Computer Engineering. His current research interests include deep learning acceleration and efficient D/CNN mapping through mapping-aware pruning in in-memory computing architectures.

Sangheum Yeon received the B.S. Degree in Electrical Engineering from Sungkyunkwan University, South Korea, in 2023, where he is currently pursuing the Ph.D. degree in Semiconductor Convergence Engineering. His current research interests include model compression through quantization and in-memory computing for efficient DNN accelerators

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with best regards

Amit Ranjan Trivedi

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with best regards

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